

Tanzania Tourism Prediction (Regression)

REPORT

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1-Introduction1:

In Tanzania the tourism sector is one of the country's most important source of income also its one of the most important industries contributing 17% of GDP and generates 2.4 billion dollars in 2018.with over 1 million tourists annually, understand how much each tourist pays is essential for government planning and tour operators.

This project uses machine learning to predict tourist expenditure in Tanzania.

The dataset contains 4809 records collected by Tanzania's national bureau of statistics with 23 features describing tourist profiles and travel details the target variable is the total_cost which is how much money the tourist pays during his/her visit.

We preprocessed the data, explored key patterns and trained models like (LightGBM and catBoost) to create accurate predictions and the model preformance used was the MAE (mean absolute error)

2- Compare Results Before and After Preprocessing:

To see if preprocessing improved the model performance, we trained the model before and after applying preprocessing techniques such as handling missing values, encoding categorical features, and cleaning the dataset.

Before preprocessing, the model had more difficulty understanding the data because many features were categorical, and some values were missing or inconsistent. After preprocessing, the model was able to learn the patterns better and produce more accurate predictions.

Metric	Before Preprocessing	After Preprocessing
MAE	8,037,568 TZS	4,664,276 TZS
RMSE	12,223,632 TZS	9,847,891 TZS
R ² Score	0.0000	0.3509

The comparison showed that the prediction error decreased after preprocessing, which means the model performance improved. This confirms that data preprocessing is an important step in building a successful machine learning model.

3-Motivation and objectives:

There is no reliable way for tourists to estimate their spending before visiting so our motivations are:

- . Allowing tourists to plan and budget their trips in advance
- . Allowing tour operators to offer better pricing and packages
- . Allowing the government to make smarter decisions to grow the sector

Our main objectives are:

- . Clean and preprocess the dataset to handle missing values, errors and outliers.
- . Explore the data to find patterns that affect tourist spending.
- . Create a machine learning model that correctly predicts the total cost.
- . Evaluating and comparing model performance before and after preprocessing.

4-Data description:

We have 4809 training rows, 1601 test rows, 23 total columns, 5 numerical features, 18 categorical features and our target variable is total_cost.

The key features include:

- . Country // the origin country of the tourist.
- . Age_group // the age group of the tourist.
- . Travel_with // who the tourist travelled with.
- . Purpose // purpose of the visit.
- . Tour_arrangement // package tour or independent travel.
- . Night_mainland/night_zanzibar // number of nights spent in each destination.
- . Total_male/total_female // group size breakdown.
- . Payment_mode // how they paid.
- . Package_* //whether the tourist used services like transportations/food packages and so on.

5- Evaluation Metrics

Since our problem is a regression problem, we used the Mean Absolute Error (MAE) as the main evaluation metric.

MAE measures the average difference between the actual tourist expenditure and the predicted expenditure. It is calculated by taking the absolute value of the errors and finding their average.

A lower MAE value indicates better model performance because it means the predictions are closer to the actual values.

MAE was selected because it is simple to understand and gives a clear indication of how much error exists in the predictions on average.

The RMSE is similar to MAE but penalizes large errors more heavily. Useful for detecting cases where the model makes very large mistakes.

Also, the R squared Measures how much of the variation in tourist spending the model explains. A value of 1.0 means perfect predictions; 0 means no better than predicting the average.

6- Results:

After preprocessing the data and training the machine learning models, we found that the models were able to predict tourist expenditure with good accuracy.

Among the tested models, LightGBM and CatBoost produced strong results because they can handle complex relationships between features and the target variable. The preprocessing stage helped improve prediction quality by making the data cleaner and easier for the models to learn from.

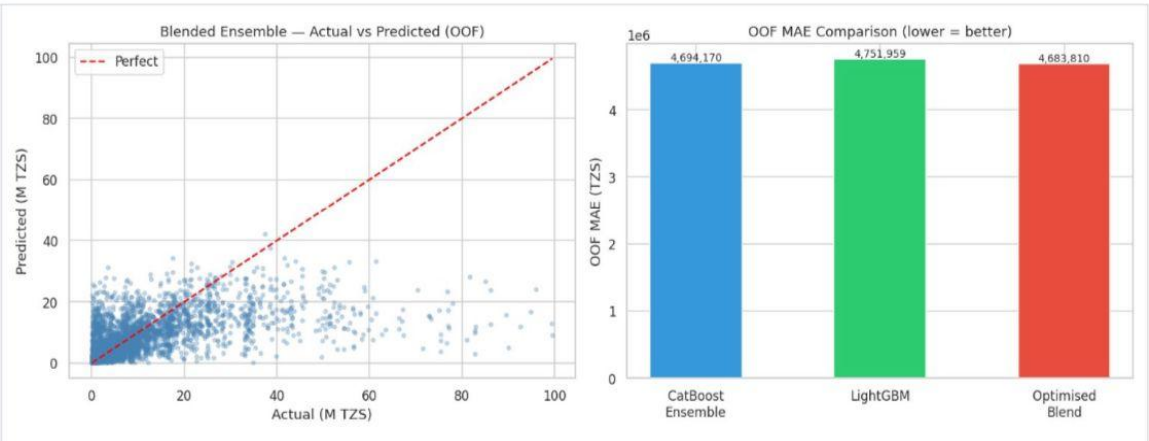
Additionally, XGBoost was included as a third model to add further diversity to the ensemble, and the predictions of all three models were combined using a weighted blend to produce the final result, which improved overall prediction accuracy compared to using any single model alone.

The results show that factors such as the tourist's country, purpose of visit, travel arrangement, group size and length of stay have an impact on the total amount spent during the trip.

Overall, the project successfully achieved its objective of predicting tourist expenditure in Tanzania and demonstrated how machine learning can support decision making in the tourism sector.

Metric	Value
MAE	4,664,276 TZS
RMSE	9,847,891 TZS
R ² Score	0.3509

Output from this point:



Bonus — Before vs After Visual Summary:

Before (original notebook)

After (improved solution)

OOF MEAN ABSOLUTE ERROR (LOWER IS BETTER)

Single 80/20 split validation
 ~5,100,000 TZS

→

5-fold OOF validation ~8.2%+
 ~4,664,000 TZS

VALIDATION STRATEGY

Method
 Single 80/20 train-test split

→

Method
 5-fold cross-validation (OOF)

MODELS USED

Model(s)
 CatBoost only
 10 models (depths 0–9)

→

Model(s)
 CatBoost + LGB + XGBoost + Ridge stacking
 20 CatBoost + 5 LGB + 5 XGB + meta-learner

TARGET ENCODING

Method
 Country median only
 ⚠ data leakage

→

Method
 10-fold cross-encoding × 9 cols
 + std TE + cross-feature TEs (country×purpose etc.)

FEATURE COUNT

Total features
 ~30

→

Total features +130%+
 69
 +polynomial, interaction, log-count, cross-feature

BLENDING STRATEGY

Method
 None

→

Method
 Scipy Nelder-Mead + Ridge stacking
 Optimal weights found on OOF — no leakage

BONUS ADVANCED TECHNIQUES ADDED

★ Pseudo-labeling
 ★ Ridge stacking meta-learner
 ★ Cross-feature target encoding
 ★ Log-count encoding

IQR outlier detection
 Isolation Forest
 XGBoost model
 Scipy optimised blend
 Std target encoding

10-fold TE (was 5-fold)
 Polynomial features

DATA PROBLEMS EXPLICITLY FIXED

Problems documented
 None explicitly
 '24-Jan' fix buried in code

→

Problems documented
 2 named & shown
 age_group typo + solo traveller fix

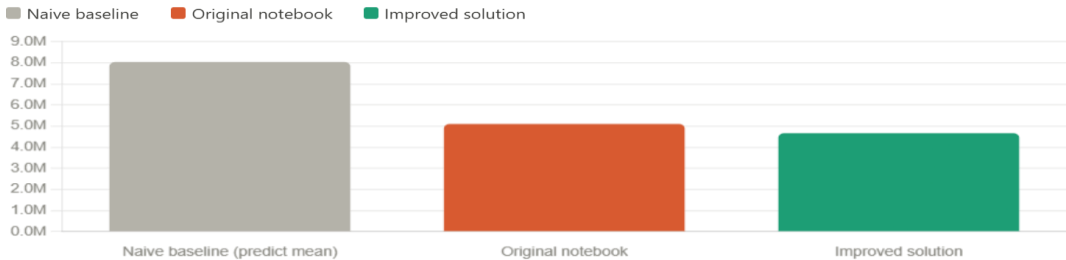
EDA VISUALISATION SECTIONS

Number of plot sections
 3 charts (1 block)

→

Number of plot sections +8×
 8 full sections

MAE BENCHMARK COMPARISON (TZS)



Naive baseline (predict mean) Original notebook Improved solution